YEAR 9 SCIENCE
PHYSICAL SCIENCES
REVISION - TEST 1

1. (a) How do the particles move in a ***transverse wave***? Give an example of this wave.

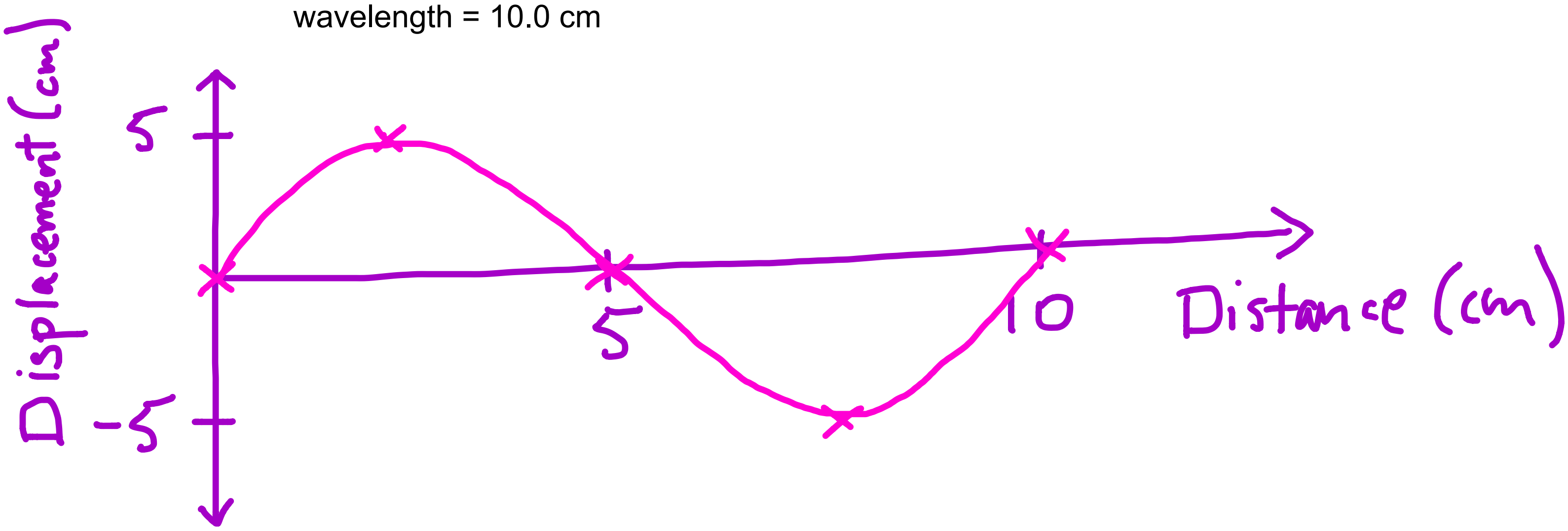
Particles move up + down - at right angles to the direction of the wave.
e.g. water waves, light

(b) How do the particles move in a ***longitudinal wave***? Give an example of this wave.

Particles move backwards + forwards - in same direction as direction of the wave.
e.g. sound

(c) Draw a displacement - distance graph to show the following. Make sure you give appropriate scales and labels.

amplitude = 5.00 cm
wavelength = 10.0 cm



2. (a) How would you describe ***light*** - what is it?

Pure energy - electromagnetic radiation.

(b) Does light need a medium (substance) to travel in? How do you know?

No - light travels in space (a vacuum) from the Sun and stars.

3. (a) List the parts of the electromagnetic spectrum in order.

radio/TV microwave infrared visible ultraviolet X-ray gamma ray

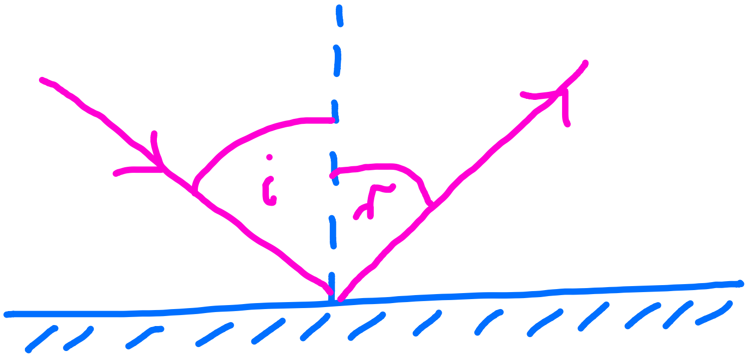
(b) Give a use or characteristic for each part.

communication cooking communication heat seeing skin cancer plants grow medical investigations cancer treatment

4. (a) What are the Laws of Reflection?

Angle of incidence = Angle of reflection

(b) Draw a simple diagram to show it.



5. What is the difference between a ***real image*** and a ***virtual image***?

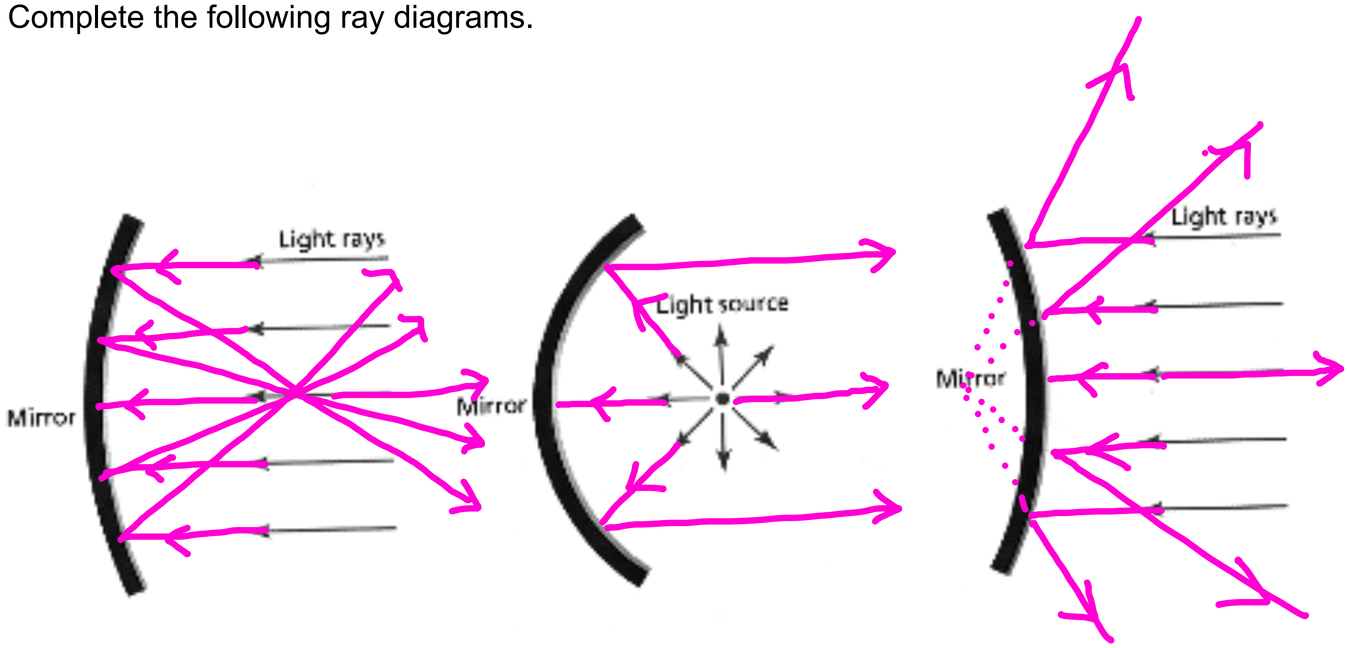
Virtual image:

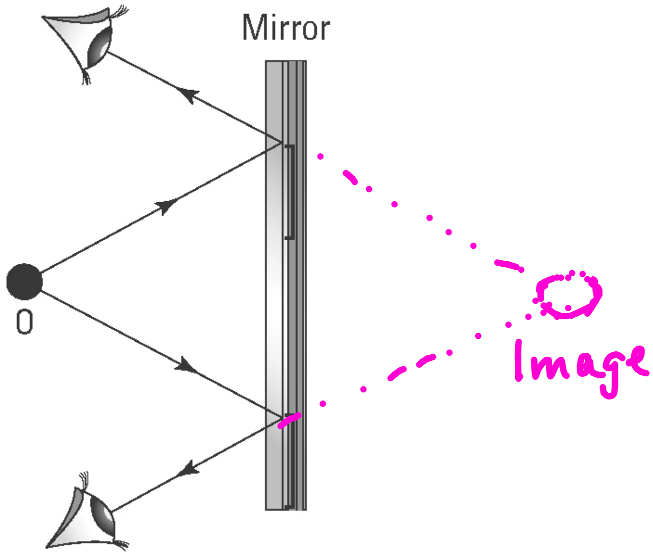
- Can't be shown on a screen.
- Rays diverge (move away) from each other.

Real image:

- Can be shown on a screen.
- Ray converge (come together) and cross each other.

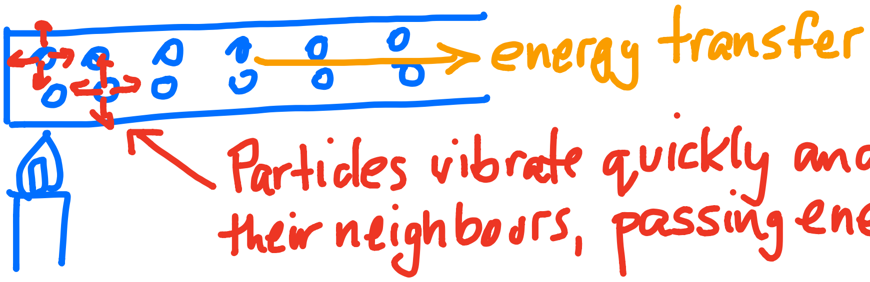
6. Complete the following ray diagrams.





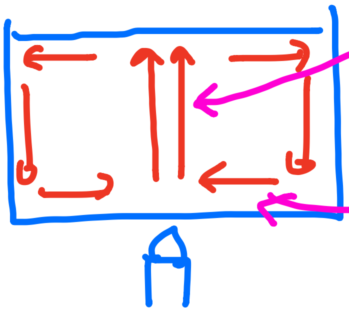
7. Explained how heat is transferred in the following.

(a) A piece of copper wire that is heated at one end.



Particles vibrate quickly and collide with their neighbours, passing energy along.

(b) A pot of warm water that is being heated.

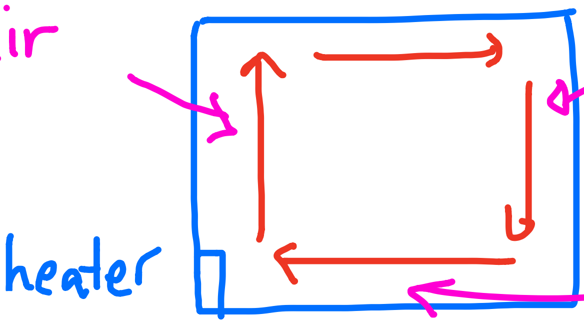


hot water becomes less dense and rises.

cool water moves in to take its place.

(c) A room with a small gas heater at floor level.

hot air rises



air cools and moves down

cool air moves in to take its place.

8. ***Metals*** have a very high thermal conductivity (<u>*able to transfer heat quickly*</u>) while ***gases*** are very low. Explain this difference.

Metals- particles close together. Small movement before colliding with the next particle.

Gases- particles very far apart so collisions do not occur easily.

9. Some very expensive cookware has copper bottoms on all the pots and pans. Suggest a reason for this.

Copper is an excellent conductor of heat.

Contents of the pot will heat faster.

10. Sports people who injure themselves are advised to put ice or a cold pack immediately on the injury to limit any internal bleeding. Why does the ice feel cold on the skin?

Heat moves from the skin (hot) to the ice (cold).

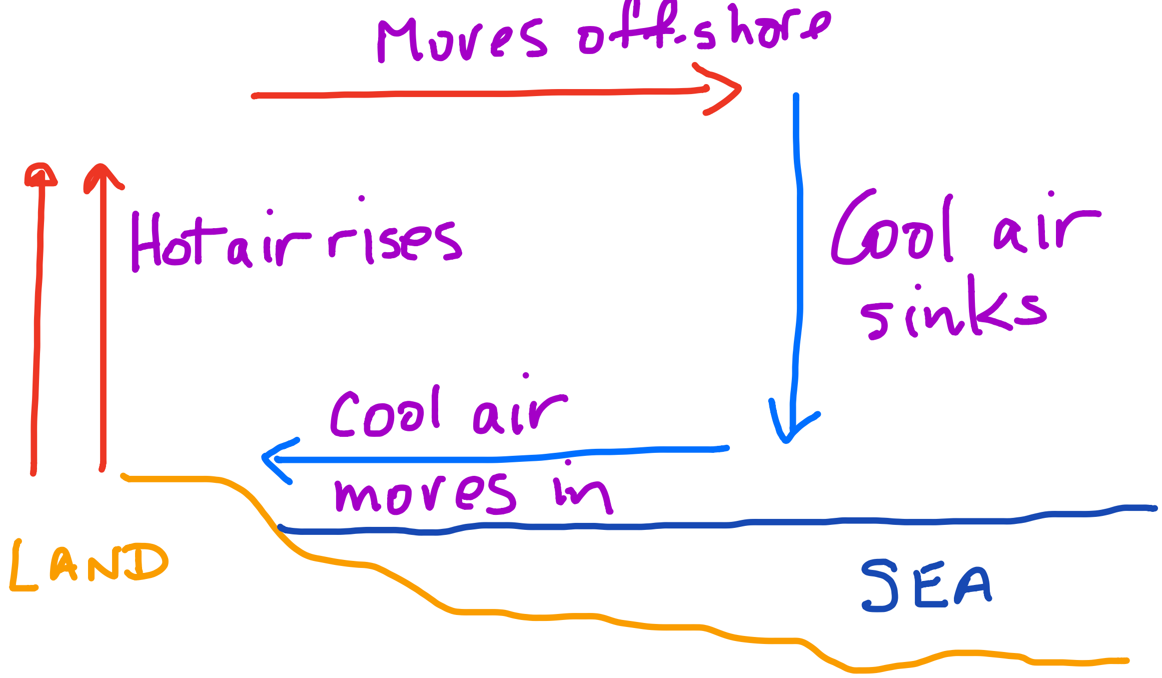
Skin goes cold.

11. Radiators on cars, trucks and other engines are _painted black_ and have _lots of very fine "fins"_ which allow the air to pass through. How do these two features make the radiator an efficient cooling device for the water carrying heat away from the engine?

- Black- radiates heat well.

- Fins - greater surface area allows more heat to leave the system.

12. Draw a simple labelled diagram to explain how a **_sea breeze_** occurs during the summer.



13. Parents become quite annoyed when one of their children stands in front of the refrigerator with the door open, thinking about what they would like to take from it. The cold air "falls" out of the fridge onto their feet and the motor starts to chill the inside again once the door closes. This costs money in the long run!

Why does the cold air "fall" out?

- Cold air is more dense than warm air.
- It sinks to the lowest level - the floor.